PATIENT CARE THEORY 2

UNIT 11, PART 1: Diabetes and Hypoglycemia

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Endocrine Emergencies

Learning Outcomes

Upon successful completion of this unit, the student will have reliably demonstrated the ability to:

- Discuss the role of endocrine glands
- Discuss the role of hormones of the endocrine system
- List the common endocrine emergencies paramedics will encounter
- Compare endocrine emergencies in terms of sensitivity and specificity of signs and symptoms
- Contrast metabolic from structural cause of altered level of awareness
- Describe and defend your treatment strategy for endocrine emergencies

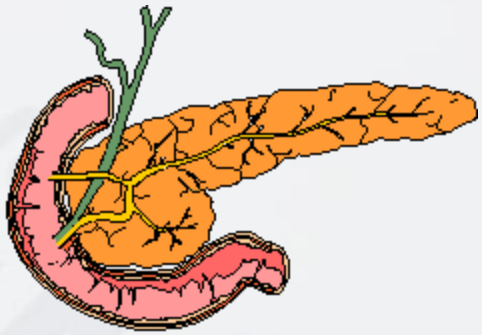
MAJOR ENDOCRINE GLANDS

- Hypothalamus
- Pituitary
- Thyroid
- Parathyroid
- Thymus
- **Pancreas**
- Adrenals
- Gonads: Ovaries/Testes

Endocrine Glands

- Secrete chemical substances (hormones) directly into the blood (ductless)
- **Hormones**: the chemical substances that control or affect processes in other glands or body systems
- Effects tend to be **systemic or widespread** (as opposed to those secreted by *exocrine glands*)
- Have a very specific effect on target tissues (which may be many or few

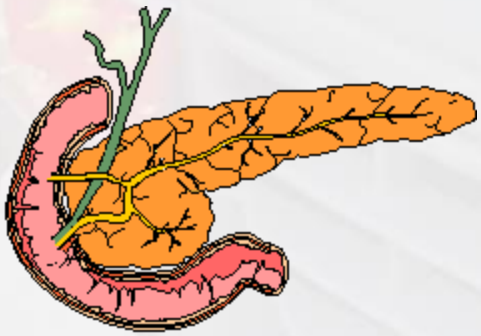
Pancreas

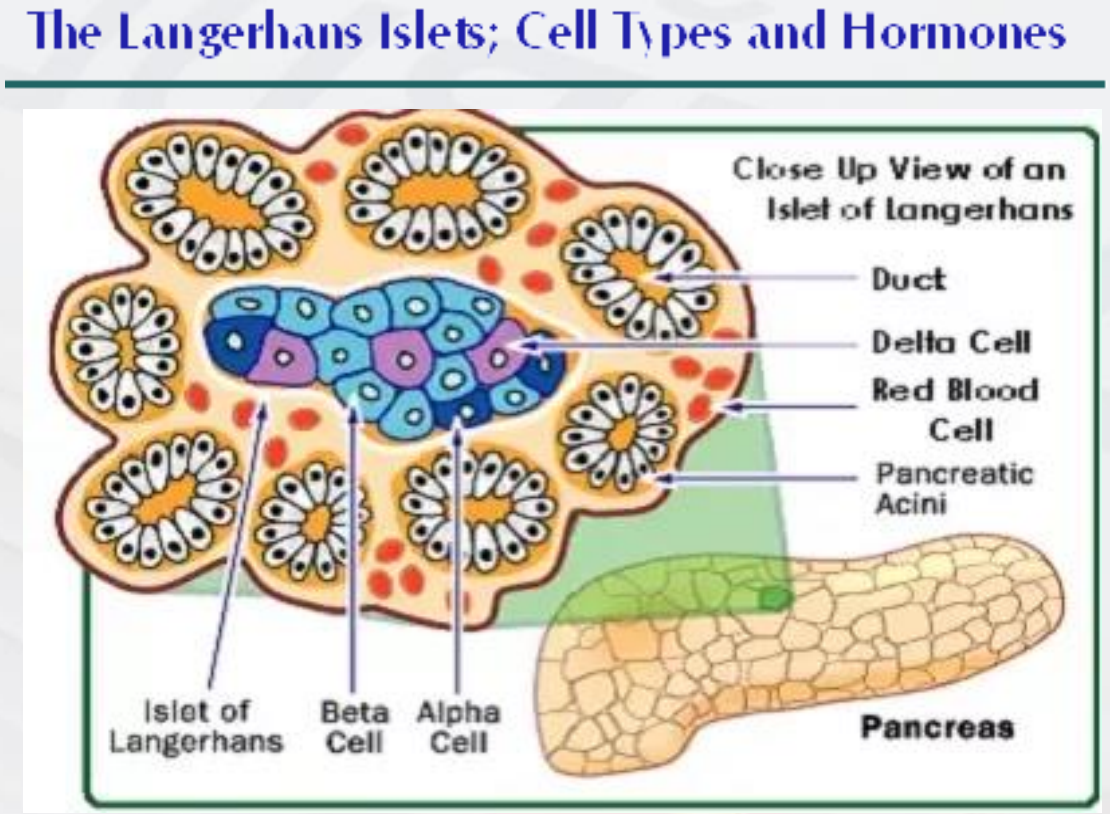


- Produces two main hormones:
  - **Insulin**
    - Produced by the beta cells in the Islets of Langerhans
    - Lowers blood glucose levels (moves glucose into the cell)
  - **Glucagon**
    - Produced and secreted by the Alpha cells of the pancreas
    - Raises blood glucose levels by breaking down glycogen to glucose
      - *Glycogen is stored primarily in the liver and skeletal muscles*
  - + Somatostatin
    - Lowers insulin or glucagon levels when those hormone levels are too high
  - Pancreas also produces digestive enzymes (exocrine function)

http://www.endocrineweb.com/diabetes/2insulin.html

Pancreas





Alpha Cells – Glucagon
Beta cells – Insulin
Delta cells - Somatostatin

Normal Values

can rise to 9 in people who've eaten desert

you can go several days without eating without BG dropping below 4 in healthy people

- Glucose is normally 4-6 mmol/L
  - 3-6 mmol/L in children < 2 years
  - 1.5 – 3 in the newborn (first 24 hours of life)
- May be high in response to catecholamine release
  - e.g. trauma, post seizure, fever, immediately after a meal, etc.
  - May also increase with infections

Insulin and Glucagon

brain, RBCs, pancreas, and liver don't need insulin for uptake

- Hormones that work to regulate blood sugar levels based on a negative feedback loop
- Insulin is produced in the Beta cells in the Islets of Langerhans in the pancreas (Islets make up ~ 5% of the pancreas)
  - It is released in response to rising glucose levels following a meal
  - Transports glucose into the cell
  - Helps to absorb amino acids and fatty acids
  - Converts excess glucose to glycogen in the liver for later use

Insulin and Glucagon

- Glucagon is a hormone produced in the Alpha cells in the pancreas
  - It is released in response to ***low*** blood glucose levels
  - Converts glycogen stores into glucose to increase the blood levels of glucose to a normal range

Diabetes

- Diabetes is marked by *inadequate insulin production*, *cellular resistance to insulin action* or both
- Insulin is needed to regulate blood sugar levels
- Insulin is the dominant hormone when glucose is high

Disorders of the Pancreas

Type 1 diabetes

- etiology: autoimmune
- an absolute or relative deficiency of insulin
- early onset (4-7 and 10-14 years of age)
- estimated 3 children out of 1000 develop IDDM by age 20 years
- previously called Juvenile Onset Diabetes
- Most children with diabetes have IDDM (Insulin Dependent Diabetes Mellitus) and a lifetime dependence on exogenous insulin

Type 1 Diabetes

- Heredity is an important factor however cause if T1D is often unclear
  - Viral infection
  - Destruction of the beta cells
- Polydipsia
- Polyuria
- Polyphagia
- Weakness
- Weight loss
- Untreated or initial presentation usually diagnosed with DKA

Disorders of the Pancreas

recent increase in early onset due to poor diets and high childhood obesity rates

- Type 2 Diabetes: onset later in life (although childhood onset is on the rise)
  - Partial destruction of insulin producing Islet cells and/or cells have a reduced response to insulin
  - sometimes referred to as age-onset diabetes
  - Typically non-insulin dependent diabetes mellitus (NIDDM)*
  - Significantly more common than Type 1 (90% of cases)
  - genetic predisposition (90% of children who develop type II have a first or second generation relative with the disease)
  - Obesity is a precipitating factor
  - Type 2 can progress to become IDDM

Long Term Complications of Diabetes

- Retinopathy
- Cataracts
- Hypertension
- Progressive renal failure
- Early coronary artery disease
- Peripheral vascular disease
- Neuropathy, both peripheral and autonomic
- Increased risk of infection

HYPOGLYCEMIA

Hypoglycemia

since brain doesn't require insulin, all you gotta do is give a shitton of sugar

- Hypoglycemia – acute onset ** ***requires prompt intervention to avoid permanent brain damage***
- Causes
  - patient took too much insulin
  - Not enough food intake
  - Exercise + not enough food
  - Oral hypoglycemics (prescription medication) + ↓ food intake
  - Alcohol consumption
  - Hormonal changes
- By far the most common diabetic emergency encountered by paramedics

metformin. it shouldn't be able to lower BG below 4 on its own

Hypoglycemia – Clinical Presentation

- Confusion, agitation → coma (from 3-14 on the GCS)
- Weakness (unilateral)
- Seizures
- Diaphoresis
- Blood glucose level (BGL) of < 4 mmol/L (adults)
  - What does the Medical Directives say for children < 2?

Hypoglycemia – Differential Diagnosis

- Hypoglycemia
  - ETOH intoxication
  - Stroke
  - CNS infection
  - Post-ictal
  - Dehydration
  - Hypothyroidism
  - Renal failure
  - Hepatic failure
  - psychosis

electrolyte imbalance causes LOA

glucagon may be ineffective, so administer d-10 instead

Key: Altered LOA, Hx, Blood glucose level **(BGL) of < 4 mmol/L (or < 3 mmol/L for age < 2),** skin PCD

Hypoglycemia – Management

- Manage the airway prn
  - Most often can be managed with jaw thrust or manual maneuvers. More invasive
- SpO_2
- O_2 prn
- Assess BGL
- ECG
- IV access & D50W/D10W

or preferred as IV is faster acting, and instantly usable by brain

- Glucagon IM

Hypoglycemia – Identification

Hypoglycemia

Age	Blood glucose level
<2 yr	<3.0 mmol/L
≥2 yr	<4.0 mmol/L

Level of Awareness (LOA)

The word 'altered' refers to a GCS that is less than normal for the patient.

The word 'unaltered' refers to a GCS that is normal for the patient. This may be a GCS <15.

Hypoglycemia – Management

Indications

Suspected hypoglycemia

Conditions

	dextrose
Age	≥ 2 years
LOA	Altered
HR	N/A
RR	N/A
SBP	N/A
Other	Hypoglycemia

	glucagon
Age	N/A (≥4 years for IN powder)
LOA	Altered
HR	N/A
RR	N/A
SBP	N/A
Other	Hypoglycemia

Hypoglycemia – Management

Contraindications

Dextrose	Glucagon
Allergy or sensitivity to dextrose	Allergy or sensitivity to glucagon
	Pheochromocytoma

adrenal tumor. induces massive release of Catecholamines, resulting in HT

Hypoglycemia – Management

Consider dextrose (if available and authorized)		
	Age	
	≥2 years	
	Concentration	**Concentration**
	10% dextrose	50% dextrose
	Route	**Route**
	IV	IV
Dose	0.2 g/kg (2 ml/kg)	0.5 g/kg (1 ml/kg)
Max. single dose	25 g (250 ml)	25 g (50 ml)
Dosing interval	10 min	10 min
Max. # of doses	2	2

Titrate dextrose to a level of awareness where the patient can safely consume complex carbohydrate.

Hypoglycemia – Management

Consider glucagon (if not using dextrose)			intranasal powder (If authorized and available)
	Age		Age
	N/A		≥4 years
	Weight	Weight	Weight
	< 25 kg	≥ 25 kg	N/A
	Route	Route	Route
	IM	IM	IN
Dose	0.5 mg	1 mg	3 mg
Max. single dose	0.5 mg	1 mg	3 mg
Dosing interval	20 min	20 min	20 min
Max. # of doses	2	2	2

Hypoglycemia – Management

Clinical Considerations

If the patient responds to dextrose or glucagon, he/she may receive oral glucose or other simple carbohydrates.

If only mild signs or symptoms are exhibited, the patient may receive oral glucose or other simple carbohydrates instead of dextrose or glucagon.

If a patient initiates an informed refusal of transport, a final set of vital signs including blood glucometry must be attempted and documented.

IV administration of dextrose applies only to PCPs authorized for PCP Autonomous IV.

Intranasal glucagon is a powder that is supplied in a commercially available single-dose intranasal device.

Treat and Discharge

Considerations for Treat and Discharge (if authorized)

All of the following criteria must be met:

- ☐ the patient is ≥18 AND <65 years old;
- ☐ the patient has a diagnosis of diabetes;
- ☐ the hypoglycemia can be explained by insulin administration with inadequate oral intake;
- ☐ the hypoglycemia promptly responded to a single administration of dextrose or glucagon as per the Medical Directive and/or consumed oral glucose or other complex carbohydrates;
- ☐ this was a single isolated episode of symptomatic hypoglycemia within the past 24 hours;
- ☐ the blood glucose is ≥4.0mmol/L after treatment;
- ☐ the patient has a return to their normal level of consciousness and is asymptomatic;
- ☐ a complete set of vital signs are within expected normal ranges;

Treat and Discharge

AND

- not an intentional overdose;
- the hypoglycemia must not be related to alcohol or substance abuse or withdrawal;
- no seizure or reported history of seizure prior to paramedic treatment,
- not on an oral hypoglycemic medication;
- hypoglycemia is not considered to be related to an acute medical illness, and;
- the patient is not pregnant.

Treat and Discharge

In addition to the above criteria, if all of the following requirements have been met, the patient can be discharged by paramedics:

- ☐ the patient has access to appropriate carbohydrates;
- ☐ a responsible adult agrees to remain with the patient for the next 4 hours;
- ☐ all of the patient or substitute decision makers questions were answered and a care plan was developed;
- ☐ the patient or substitute decision maker has been advised to follow up with their primary health care team or provider;
- ☐ clear instructions to call 911 were provided should symptoms redevelop;
- ☐ patient or substitute decision maker has the ability to access 911 should symptoms redevelop, and;
- ☐ patient or substitute decision maker consents to the discharge.

Patch to BHP if you are unclear whether the patients meets all the discharge criteria

Pheochromocytoma

- Rare catecholamine secreting tumor on the adrenal gland (0.05-0.2% of hypertensive individuals)*
- Clinical presentation: hypertensive crisis, headache, altered LOA, diaphoresis, palpitations, tremor, nausea, weakness, anxiety, sense of doom, epigastric pain, flank pain, constipation, weight loss, hyperglycemia, hypoglycemia
- **Glucagon is Contraindicated**
  - May worsen hypoglycemia
  - Cause hypertensive crisis/stroke

Insulinoma

- A rare neuroendocrine tumor deriving mainly from pancreatic islet cells that produce excessive amounts of ***insulin***
- about 90% of insulinomas are benign
- Symptoms: episodic hypoglycemia
- central nervous system (CNS) dysfunction temporally related to hypoglycemia (confusion, anxiety, stupor, convulsions, coma)
- Diagnosis can be difficult
  - Primary treatment is surgical removal

QUESTIONS